

Austin Pounder

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in LinkedIn

G Scholar

Research Interests

Organic Chemistry

Catalysis

Reaction Mechanisms

Methodology Development

Synthesis

Nucleic Acids

Photochemistry

Fluorescence

Physical Organic Chemistry

(TD)-DFT

MD Simulations

Digital Chemistry

Academic Appointments

- **NSERC Canada Postdoctoral Research Award** May 2026 – April 2028
Department of Chemistry, Colorado State University, USA
Advisor: Prof. Robert S. Paton
- **Alberta Innovates Postdoctoral Fellowship** May 2024 – April 2026
Department of Chemistry and Biochemistry, University of Lethbridge, Canada
Advisor: Prof. Stacey D. Wetmore Co-advisor: Dr Jean-Denys Hamel

Education

- **PhD in Experimental and Computational Organic Chemistry** Sept. 2019 – April 2024
Guelph-Waterloo Center for Graduate Work in Chemistry and Biochemistry (GWC²),
Guelph Campus
Advisor: Prof. William Tam Co-advisor: Prof. Leanne D. Chen
Thesis: "Experimental and Theoretical Investigations in Transition Metal Catalysed Reactions of Heterobicyclic Alkenes"
- **BSc in Biological and Pharmaceutical Chemistry With Distinction** Sept. 2015 – April 2019
Department of Chemistry, University of Guelph
Advisor: Prof. William Tam
Thesis: "Acid-Catalyzed Intramolecular Ring-Opening Reactions of Cyclopropanated Oxabenzonorbornadienes"

Research Experience

- **Visiting Researcher** Nov. 2024 – Jan. 2025
Department of Chemistry, University of Guelph
Supervisor: Prof. Richard A. Manderville
 - DNA synthesis and on-strand fluorophore synthesis.
 - Fluorophore synthesis and photophysics studies focusing on photobasicity.
- **Visiting Researcher** Sept. 2023 – Nov. 2023
Department of Chemistry, University of Lethbridge
Supervisor: Prof. Stacey D. Wetmore
 - Conducted an evaluation of TD-DFT computed absorption/emission spectra using Gaussian and ORCA.
 - Identified optimal methods for dye calculations; performed MD simulations in AMBER to forecast dye-DNA photophysical behavior for biosensing applications.
- **Laboratory Research Assistant** May 2019 – Aug. 2019
Department of Chemistry, University of Guelph
Supervisor: Prof. William Tam
 - Developed novel 1,4-disubstituted oxabicyclic alkenes.
 - Optimized Ni-catalyzed ring-opening reactions with C1-tethered aryl halides.
- **Senior Research Project Student** Sept. 2018 – April 2019
Department of Chemistry, University of Guelph
Supervisor: Prof. William Tam
 - Synthesized novel 1,5-disubstituted cyclopropanated oxabenzonorbornadienes.
 - Investigated acid- and Ir-catalyzed ring-opening with tethered nucleophiles and water.

- **Laboratory Research Assistant**
Department of Chemistry, University of Guelph
Supervisor: Prof. Adrian Schwan
- Synthesized naphthyl- and anthracenyl-sulfenates.
- Explored diastereoselective induction using chiral electrophiles.

May 2018 – Aug. 2018

Honours and Awards

National/International

• 2026	NSERC Canada Postdoctoral Research Award (NSERC CPRA)	\$140,000
• 2024	Alberta Innovates Postdoctoral Fellowship (AI-PDF)	\$140,000
• 2022	Canadian Chemistry Conference – Organic Division Poster Winner	\$1,000
• 2021	NSERC Postgraduate Scholarship – Doctoral (NSERC PGS-D)	\$63,000
• 2020	NSERC Canada Graduate Scholarship – Master's (NSERC CGS-M)	\$17,500
• 2018	NSERC Undergraduate Student Research Award (NSERC USRA)	\$6,000

Provincial

• 2021	Ontario Graduate Scholarship (OGS)	\$15,000
• 2020	Ontario Graduate Scholarship (OGS)	\$15,000

Institutional

• 2025	Postdoctoral Research Dissemination Travel Award	\$1,000
• 2023	GWC ² AGM – Best Poster Winner	\$200
• 2022	Charles S. Humphrey Graduate Fellowship in Chemistry	\$2,000
• 2022	CEPS Graduate Dean's Scholarship	\$3,500
• 2021	Charles S. Humphrey Graduate Fellowship in Chemistry	\$2,000
• 2019	UoG Graduate Tuition Scholarship (GTS)	\$16,000
• 2017	Undergraduate Chemistry Award – John Wiley & Sons, Inc. Book Prize	
• 2015	Undergraduate Dean's Honours List	
• 2015	Undergraduate Entrance Scholarship	\$2,000
• 2015	James R. Hoffa Memorial Scholarship	\$13,200

Publications

Publication Highlights (Co)-First Authorship = 15 HQP Authorship = 10 Corresponding Author = 3

* denotes corresponding author, — denotes HQP trained by A. Pounder, ‡ denotes co-first author

Manuscript in Preparation

40. **Pounder, A.**, Wetmore, S. D.* (2026). From Dynamics to Design: Computational Insights into Fluorogenic RNA Aptamers. *In preparation*. PDF work
39. **Pounder, A.**, Sadiq, F., Wetmore, S. D.* (2026). A DFT Investigation Into the Impact of Post-Transcriptional Modifications in the Anticodon Loop of tRNA on Stacking Interactions. *In preparation*. PDF work
38. **Pounder, A.**, Yaqoob, U., Olvis, Z., Lea, M., Kothe, U., Wetmore, S. D.* (2026). Simulation Criteria for Structural Convergence on RNA: A Case Study of tRNA. *In preparation*. PDF work
37. Tajudeen, R., Quinn, T., **Pounder, A.**, Wetmore, S. D.* (2026). Impact of 2'-Ribose Modification Pattern on the Structure of siRNA Therapeutics: A Molecular Dynamics Simulation Study. *In preparation*. PDF work
36. Manoj, N.; **Pounder, A.**; Paton, R. S.* Machine Learned Interatomic Potentials for Organic Selectivity. [PDF work](#)

Submitted

35. **Pounder, A.**, Wetmore, S. D.* (2026). The Performance of Time-Dependent Density Functional Theory for Electronic Excitations of Nucleic Acid Fluorescent Probes. *J. Chem. Theory Comput.* [145543F3-E71D-47E1-B681-1389C63E2C3C]. PDF work
34. Pavlovic, M., **Pounder, A.**, Varnovitski, N.; Wetmore, S. D.*, Manderville, R. M.* (2026). Benzimidazole Modification of DNA Aptamers for Ultra-Responsive Protein-Induced Fluorescence Enhancement. *Anal. Chem.* [ac-2026-05353j] PDF work

33. Saraya, J. S.; **Pounder, A.**; Backhouse, M. R.; Wetmore, S. D.; O'Flaherty, D. K.* (2026). Universal Solid Support Compatibility with Amino-Modified Oligonucleotide Synthesis. *Chem. Sci.* [SC-EDG-06-2026-005141]. PDF work
32. Li, Y.; **Pounder, A.**; Chokkapu, E. R.; Oh, J.; Gowda, R. R.; Casaday, C. E.; Chen, E. Y.-X.; Paton, R. S.*; Cundari, T. R.*; Dong, Y.* (2026). Spin-State Suppression of β -Hydride Elimination En Route to Kinetically Stabilized High-Spin Fe-Alkyl Complexes Exhibiting Broad Reactivity in Small-Molecule and Polymerization Catalysis. *Chem. Catal.* [CHEM-CATALYSIS-D-26-00192]. PDF work

Published

31. Roby, D. J., **Pounder, A.**, Wetmore, S. D.*, Manderville, R. A.* (2026). Isomeric Indole-Chalcone Surrogates as Complementary Reporters of Nucleic Acid Binding Thermodynamics. *J. Phys. Chem. B* accepted. PDF work
30. Dudas, T.[‡], **Pounder, A.**[‡], Umeris, I. M., Wetmore, S. D.*, Hamel, J-D.* (2026). An Experimental and Computational Investigation into the Allylboration/Defluorinative Semipinacol Rearrangement Cascade Reaction. *J. Org. Chem.* 91, 7730–7737. PDF work
29. **Pounder, A.**, Lilienthal, E., Solorzano, J., Wetmore, S. D.*, Hamel, J-D.* (2026). Structural Characterization of Difluoroglycine Betaines. *J. Mol. Struct.*, 1368, 146292. PDF work
28. Semeniuk, T., **Pounder, A.**, Toporkov, D., Wetmore, S. D.*, Hamel, J-D.* (2025). Chemoselective Divergence via Substrate Control: Decarboxylative and Decarbonylative Coupling of 2-Oxoacids with Allylic Difluorides. *Eur. J. Org. Chem.*, e202501058. PDF work
27. Neilson, S., **Pounder, A.**, Thwaites, A. E., Wetmore, S. D.*, Hamel, J-D.* (2025). Decarboxylative Redox Bicyclo[1.1.1]pentylation: Benchmarking the Influence of Backbone Substituents on C–C Bond-Forming Reactions. *Org. Lett.*, 27, 13740–13744. PDF work
26. **Pounder, A.**, Pavlovic, M., Chow, D., Regan, K. T., Chen, A.*, Manderville, R. M.*, Wetmore, S. D.* (2025). Photobasicity-Triggered Twisted Intramolecular Charge Transfer of Push-Pull Chromophores. *J. Phys. Chem. B*, 129, 8485–8493. Invited contribution to “Jacek Lipkowski Festschrift” Special Issue. PDF work
25. Murray, M. T., **Pounder, A.**, Johnson, R. E., Regan, K. T., Manderville, R. A., Wetmore, S. D.* (2025). A Multiscale Computational and Experimental Study of TBA-Fluorescent Probes for Protein Sensing: Photobasicity Over Twisted Intramolecular Charge Transfer as a New Mechanism for Protein Induced Fluorescence Enhancement. *Phys. Chem. Chem. Phys.*, 27, 17864–17876. Contribution to “Showcasing the Breadth of Physical Chemistry Chemical Physics Research in Canada” and “Celebrating International Women’s day 2026: Women in physical chemistry” Thematic Issues. PDF work
24. **Pounder, A.***, King, N., Tam, W.* (2025). Ruthenium-Catalyzed Isomerization of Oxabicyclic Alkenes to 1, 2-Naphthalene Oxides: A Density Functional Theory Study *Eur. J. Org. Chem.*, 28, e202500192. PhD work
23. Regan, K. T.[‡], **Pounder, A.**[‡], Johnson, R. E., Murray, M. T., Glowacki, H. X., Wetmore, S. D.*, Manderville, R. A.* (2025). Modular Access to Nucleobase GFP-Surrogates: Smart Probes for Ratiometric Nucleic Acid Diagnostics. *Chem. Sci.*, 16, 6468–6479. PDF work
22. Johnson, R. E., **Pounder, A.**, van der Zalm, J., Chen, A.*, Bell, I., Van Raay, T.*, Wetmore, S. D., Manderville, R. M.* (2024). Thieno[3,2-*b*]thiophene for the Construction of Far-Red Molecular Rotor Hemicyanines as High-Affinity DNA Aptamer Fluorogenic Reporters. *Anal. Chem.*, 96, 16252–16259. PDF work
21. **Pounder, A.**, Neufeld, E., Chen, L. D.*, Tam, W.* (2024). Nickel-Catalyzed Arylation/Lactonization Cascade Reactions for the Synthesis of Bicyclo-[3.2.1]-Lactones. *Organometallics*, 43, 2443–2449. PhD work
20. **Pounder, A.**[‡], Neufeld, E.[‡], Chen, L. D.*, Tam, W.* (2024). Rhodium-Catalyzed Ring-Opening Reactions of Heterobicyclic Alkenes with Heteroarene Nucleophiles: an Experimental and Computational Investigation *Can. J. Chem.*, 102, 696–705. Contribution to “Celebrating 150 Years of Chemistry at the University of Guelph” Special Issue. PhD work
19. **Pounder, A.***, van der Zalm, J., Chen, A., Tam, W.* (2024). Cobalt-Catalyzed Hydrogenation and Deuteration of Bicyclic Alkenes in Aqueous Environment. *Eur. J. Org. Chem.*, e202301306. Highlighted in SynFacts. PhD work
18. Regan, K. T., **Pounder, A.**, Lin, C., Chen, L. D.*, Manderville, R. A.* (2023). Isomer-Specific Solvatochromic and Molecular Rotor Properties of ESIPT-Active Push–Pull Fluorescent Chalcone Dyes *J. Phys. Chem. A*, 127, 8365–8373. PhD work
17. Neufeld, E., **Pounder, A.**, Tam, W.* (2023). Rhodium-Catalyzed Ring-Opening Reactions of Heterobicyclic Alkenes with Heteroarene Nucleophiles. *Tetrahedron Lett.*, 127, 154685. PhD work
16. **Pounder, A.***, Neufeld, E., Myler, P., Tam, W. (2023)* Transition-Metal-Catalyzed Domino Reactions of Strained Bicyclic Alkenes. *Beilstein J. Org. Chem.*, 19, 487–540. Invited contribution to “Catalytic multi-step domino and one-pot reactions” Thematic Issue. PhD work

15. **Pounder, A.**, Farkas, M., Chen, L. D.*, Tam, W.* (2023). Iridium/Zinc-Co-Catalyzed Ring-Opening Reactions of Oxabicyclic Alkenes with Indole Nucleophiles: A Combined Experimental and Theoretical Study *Organometallics*, 42, 780–792. PhD work
14. **Pounder, A.**[‡], Ho, A.[‡], Kiran, V., Chen, L. D., Tam, W.* (2022). Iridium-Catalyzed Hydroacylation Reactions of C1-Substituted Oxabenzonorbornadienes with Salicylaldehyde: an Experimental and Computational Study *Beilstein J. Org. Chem.*, 18, 251–261. PhD work
13. Ho, A., **Pounder, A.**, Koh, S., Macleod, M., Carlson, E., Tam, W.* (2022). Acid-Catalyzed Intramolecular Ring-Opening Reactions of Cyclopropanated Oxabenzonorbornadienes with Carboxylic Acid Nucleophiles *Synthesis*, 54, 1422–1430. BSc work
12. **Pounder, A.**, Tam, W.* (2021). Iron-Catalyzed Domino Coupling of π -Systems. *Beilstein J. Org. Chem.*, 17, 2848–2893. PhD work
11. **Pounder, A.**, Tam, W., Chen, L. D.* (2021). The Mechanism and Origin of Enantioselectivity in the Rhodium-Catalyzed Asymmetric Ring-Opening Reactions of Oxabicyclic Alkenes with Organoboronic Acids: A DFT Investigation *Organometallics*, 40, 1588–1597. PhD work
10. **Pounder, A.**, Bishop, F., Chen, L. D.*, Tam, W.* (2021). A DFT Study on the Mechanism and Origin of Regioselectivity in the Rhodium/Diene-Catalyzed Ring-Opening Reactions of C1-Substituted Oxabenzonorbornadienes with Arylboronic Acids *Eur. J. Org. Chem.*, 1901–1908. PhD work
9. **Pounder, A.**, Ho, A., Macleod, M., Tam, W.* (2021). Chemistry of C1-Substituted Oxabenzonorbornadienes. *Curr. Org. Synth.*, 18, 446–474. PhD work
8. **Pounder, A.**, Chen, L. D., Tam, W.* (2021). Ruthenium-Catalyzed [2+2] versus Homo Diels–Alder [2+2+2] Cycloadditions of Norbornadiene and Disubstituted Alkynes: A DFT Study *ACS Omega*, 6, 900–911. PhD work
7. Koh, S., **Pounder, A.**, Brown, E., Tam, W.* (2020). Intramolecular Nickel-Catalyzed Ring-Opening Reactions of Oxabenzonorbornadienes with C1-Tethered Aryl Halides: An Improvement of Method *Eur. J. Org. Chem.*, 4558–4562. PhD work
6. Koh, S., **Pounder, A.**, Brown, E., Tam, W.* (2020). Intramolecular Palladium-Catalyzed Ring Opening of Oxabenzonorbornadienes with C1-Tethered Aryl Halides. *Org. Lett.*, 22, 3433–3437. PhD work
5. Lough, A. J.*, **Pounder, A.**, Wicks, C., Tam, W. (2020). 1,2,3,5-Tetrahydronaphtho[2,1-c]oxepine. *IUCrData*, 5, x200288. BSc work
4. Lough, A. J.*, **Pounder, A.**, Tam, W. (2020). 12-Ethyl-6a,10a-dihydro-5H-6-oxachrysene. *IUCrData*, 5, x200286. PhD work
3. Wicks, C., Koh, S., **Pounder, A.**, Carlson, E., Tam, W.* (2019). Acid-catalyzed intramolecular ring-opening reactions of cyclopropanated oxabenzonorbornadienes with alcohol nucleophiles. *Tetrahedron Lett.*, 60, 151228. BSc work
2. Petko, D., **Pounder, A.**, Tam, W.* (2019). Ruthenium-Catalyzed [2+2+2] Bis-Homo-Diels–Alder Cycloadditions of 1, 5-Cyclooctadiene with Alkynyl Phosphonates. *Synthesis*, 51, 4271–4278. BSc work
1. Hill, J., Wicks, C., **Pounder, A.**, Tam, W.* (2019). Iridium-Catalyzed Ring-Opening of Unsymmetrical Oxabenzonorbornadienes. *Tetrahedron Lett.*, 60, 150990. BSc work

Current Projects

- **Pounder, A.**; Quinn, T.; Gu, E.; Wetmore, S. D.*. Deep Learning RNA Structure. [PDF work](#)
- Capperault, M. **Pounder, A.**; Dupradeau, F-Y*, Wetmore, S. D.*, O’Flaherty, D.* Amino-Substituted DNA Oligomers. [PDF work](#)
- Li, L.; **Pounder, A.**; Paton, R. S.*; Rawal, V. H.* Diastereoselective α -Vinylolation of Chiral Oppolzer’s sultams Enabled by 9-Bismatriptycene (BisTrip). [PDF work](#)
- Li, L.; **Pounder, A.**; Paton, R. S.*; Rawal, V. H.* Enantioselective α -Arylation of Enolates Enabled by Chiral 9-Bismatriptycene (BisTrip). [PDF work](#)
- Shin, Y-J.; Li, L.; **Pounder, A.**; Paton, R. S.*; Rawal, V. H.* Transition-Metal Free Sonogashira-type Coupling Enabled by a Hypervalent Bismuth Reagent. [PDF work](#)
- Salazar-Barrientos, T. V.; **Pounder, A.**; Wetmore, S. D.; Hayes, P. G.* Benchmarking Computational Techniques for the Geometry Optimization of Transition-Metal Phosphasalen Complexes. [PDF work](#)

Talks & Conference Presentations

Oral Presentations

9. **(Invited)** University of Lethbridge Chemistry Seminar Series, Lethbridge, 2026. “Experimenting with Chemistry” [PDF work](#)
8. 108th Canadian Chemistry Conference and Exhibition (CCCE/CSC), Ottawa, 2025. “Computational Insights into Modular Nucleobase GFP-Surrogates.” [PDF work](#)
7. **(Invited)** Canadian Centre for Research in Advanced Fluorine Technologies (C-CRAFT) and University of Lethbridge Department of Chemistry and Biochemistry Seminar Series, Lethbridge, 2023. “From Theoretical Insights to Synthetic Utility: Transition-Metal Catalyzed Transformations of Heterobicyclic Alkenes.” [PhD work](#)
6. 106th Canadian Chemistry Conference and Exhibition (CCCE/CSC), Vancouver, 2023. “Iridium- and Rhodium-Catalyzed Ring-Opening Reactions of Heterobicyclic Alkenes with Electron-Rich Arenes.” [PhD work](#)
5. **(Invited)** GWC² Seminar Series, Guelph, 2022. “Combined Experimental and Theoretical Studies of Iridium Catalyzed Reactions of Heterobicyclic Alkenes.” [PhD work](#)
4. **(Invited)** CIC Canadian Organic Invitational (CORGI) Virtual Seminar Series, Online, 2022. “Nickel- and Palladium-Catalyzed Intramolecular Arylative Ring-Opening Reactions of Oxabicyclic Alkenes.” [PhD work](#)
3. 31st Quebec-Ontario Mini-Symposium for Synthetic and Bioorganic Chemistry (QOMSBOD), Online, 2021. “Intramolecular Nickel- and Palladium-Catalyzed Ring-Opening Reactions of Oxabicyclic Alkenes with C1-Tethered Aryl Halides.” [PhD work](#)
2. 24th Chemistry and Biochemistry Graduate Research Conference (CBGRC), Online, 2021. “Intramolecular Nickel- and Palladium-Catalyzed Ring-Opening Reactions of Oxabicyclic Alkenes with C1-Tethered Aryl Halides.” [PhD work](#)
1. 47th Southern Ontario Undergraduate Student Chemistry Conference (SOUSCC), Toronto, 2019. “Acid-Catalyzed Intramolecular Ring-Opening Reactions of 1,5-Disubstituted Cyclopropanated Oxabenzonorbornadiene.” [BSc work](#)

Poster Presentations

9. 18th Annual Chinook Symposium for Chemistry and Biochemistry, Lethbridge, 2025. “Computational Insights into Modular Nucleobase GFP-Surrogates.” [PDF work](#)
8. 13th Triennial Congress of the World Association of Theoretical and Computational Chemists (WATOC), Oslo, Norway, 2025. “Computational Insights into Modular Nucleobase GFP-Surrogates.” [PDF work](#)
7. 108th Canadian Chemistry Conference and Exhibition (CCCE/CSC), Ottawa, 2025. “Computational Insights into Defluorinative Semipinacol Rearrangements.” [PDF work](#)
6. RiboWest 2025, Lethbridge, 2025. “From Dynamics to Design: Computational Insights into Fluorogenic RNA Aptamers.” [PDF work](#)
5. 16th Annual Chinook Symposium for Chemistry and Biochemistry, Lethbridge, 2023. “Rhodium-Catalyzed Ring-Opening Reactions of Heterobicyclic Alkenes with Electron-Rich Arenes.” [PhD work](#)
4. GWC² Annual General Meeting, Guelph, 2023. “Rhodium-Catalyzed Ring-Opening Reactions of Heterobicyclic Alkenes with Electron-Rich Arenes.” [PhD work](#)
3. **(Winner – Organic Division)** 105th Canadian Chemistry Conference and Exhibition (CCCE/CSC), Calgary, 2022. “Intramolecular Nickel- and Palladium-Catalyzed Ring-Opening Reactions of Oxabicyclic Alkenes with C1-Tethered Aryl Halides.” [PhD work](#)
2. 103rd Canadian Chemistry Conference and Exhibition (CCCE/CSC), Online, 2020. “Iridium-Catalyzed Ring-Opening Reactions of Unsymmetrical Oxabenzonorbornadienes with Water and Alcohol Nucleophiles.” [BSc work](#)
1. College of Engineering Physical Sciences Undergraduate Poster Session – University of Guelph, Guelph, 2018. “Exploring Diastereoselective Sulfenyl Alkylations with Chiral Electrophiles.” [BSc work](#)

Peer-Reviewing Participation

Summary of participation in peer-reviewing journal articles

2022–Present

Format: Journal name (number of reviews), total count 6.

- *Nature Chemistry* (1)
- *Pure and Applied Chemistry* (1)
- *Asian Journal of Organic Chemistry* (1)

- *Chemistry Select (1)*
- *ACS Catalysis (1)*

- *Journal of Organic Chemistry (1)*

Leadership, Outreach, and Volunteering

- **2026** *Meeting of the Minds Judge*
Acted as poster and oral judge at the Meeting of the Minds Conference
- **2025** *Chinook Symposium Judge*
Acted as a poster judge at the 18th Annual Chinook Symposium for Chemistry and Biochemistry.
- **2025** *CSC 2025 Judge*
Acted as a poster judge at the 2025 Canadian Chemistry Conference and Exhibition for both the Physical, Theoretical and Computational Chemistry (PTC) and Biological and Medicinal Chemistry Division (BM).
- **2025** *RiboWest 2025 Judge*
Acted as a speaker judge at the RiboWest 2025 conference.
- **2025** *Women in STEM Conference Judge*
Acted as a poster judge at the 3rd Annual Women in STEM Conference.
- **2024, 2025** *Graduate Internal Award Review Committee*
Reviewed and ranked internal graduate scholarship applications for the Science/Health committee.
- **2024** *Indigenous STEM Recruitment*
Participated in a departmental initiative to brainstorm strategies for recruiting Indigenous students into independent studies and honours theses in Chemistry and Biochemistry.
- **2018–2024** *Guelph Chemistry Department Outreach Programs*
Demonstrated labs to high school students visiting the university. Participated in outreach events including “Let’s Talk Science!” and the “Science Olympics.” As Vice-President Internal of the Graduate Chemistry Society, coordinated outreach visits to local high schools to share chemistry experiments and personal academic experiences.
- **2022–2023** *Vice President Internal, Graduate Chemistry Society*
Co-founded and organized activities for the graduate chemistry club. Led fall orientation events, departmental socials, and regular monthly meetings. Coordinated community-building events such as bowling nights and holiday celebrations.
- **2023** *Chinook Symposium Judge*
Poster judge at the 16th Annual Chinook Symposium for Chemistry and Biochemistry.
- **2022** *Conference Organizer*
Supported planning and execution of the 29th International Symposium on the Organic Chemistry of Sulphur (ISOCS-29). Managed abstract submissions, registration, and provided on-site assistance.

Teaching

Teaching Experience

- **Chemistry Sessional Lecturer**, Dept. of Chemistry and Biochemistry, University of Lethbridge
 - CHEM*3070 – Advanced Physical Chemistry Fall 2025
Delivered instruction in a computational chemistry laboratory course, incorporating applied computation for theoretical chemistry and introductory ab initio quantum chemical methods.
- **Chemistry Teaching Assistant**, Dept. of Chemistry, University of Guelph
 - CHEM*2720 / CHEM*2700 – Fundamental Organic Chemistry/Organic Chemistry I Winter 2024
Designed and graded assignments and exams. Held weekly tutorials and seminars.
 - CHEM*2720 – Fundamental Organic Chemistry Summer 2023
Designed and graded assignments and exams.
 - CHEM*2720 – Fundamental Organic Chemistry Winter 2023
Assisted in assignment design and grading.
 - CHEM*3750 – Organic Chemistry II Fall 2022
Supervised wet labs. Graded assignments and exams.
 - CHEM*2720 – Fundamental Organic Chemistry Summer 2022
Assisted in assignment design and grading.
 - CHEM*2720 – Fundamental Organic Chemistry Winter 2022
Assisted in assignment design and grading.

- CHEM*3750 – Organic Chemistry II Fall 2021
Supervised wet labs. Graded assignments and exams.
- CHEM*2720 – Fundamental Organic Chemistry Summer 2021
Assisted in assignment design and grading.
- CHEM*4730 – Synthetic Organic Chemistry Fall 2020
Designed assignments. Held tutorials. Graded assignments and exams.
- CHEM*2700 – Organic Chemistry I Winter 2020
Supervised wet labs. Graded assignments and exams.
- CHEM*1040 – General Chemistry I Fall 2019
Supervised wet labs. Graded assignments and exams.

HQP Supervision

All HQP worked on projects I led. Each was assigned a subsection of a research project under my mentorship. 4th-year project students received additional oral and report-writing guidance.

Supervision of Experimental and Computational Work

- Eric Neufeld (MSc - UoG) (currently: PhD student at U. of Ottawa) Apr 2022 – May 2024
Rhodium-catalyzed ring-opening reactions of heterobicyclic alkenes.
- Peter Myler (4th year undergraduate - UoG) (currently: MSc student at McMaster U.) Apr 2022 – Apr 2023
Palladium-catalyzed hydroformylation reactions.
- Taylor Rounds (4th year undergraduate - UoG) (currently: QC chemist at Kenvue) Jan 2020 – Dec 2020
Palladium/nickel-catalyzed arylations and acid-catalyzed ring-openings.
- Casandra Rooke (4th year undergraduate - UoG) (currently: Teacher) Jan 2020 – Dec 2020
Nickel-catalyzed [2+2] cycloadditions of alkynyl phosphonates.
- Megan Farkas (4th year undergraduate - UoG) (currently: QC chemist at Kenvue) Jan 2020 – Dec 2020
Rhodium-catalyzed ring-opening reactions with indoles.
- Matthew Macleod (4th year undergraduate - UoG) (currently: Music producer) May 2019 – Aug 2019
Acid-catalyzed ring-opening reactions.

Supervision of Computational Work

- Eugene Gu (3rd year undergraduate - UoL) (currently: BSc student at U. of Lethbridge) Sept 2025 – Dec 2025
Machine learning of RNA structure.
- Umer Yaqoob (PhD - UoL) (currently: PhD student at U. of Lethbridge) May 2025 – April 2026
MD simulations and criteria for structural convergence on RNA.
- Ridwan Tajudeen (PhD - UoL) (currently: PhD student at U. of Lethbridge) May 2025 – April 2026
MD simulations on siRNA drugs.
- Khushi Patel (High-school - UoL) (currently: Grade 12 high school student) July 2025 – Aug 2025
MD simulations on siRNA drugs.
- Shivam Tikoo (Visiting PhD - UoL) (currently: PhD student at U. of Concordia) July 2025 – Aug 2025
MD simulations and QM calculations on Se-tethered DNA.
- Tyler Quinn (4th year undergraduate - UoL) (currently: BSc student at U. of Lethbridge) May 2025 – April 2026
MD simulations on siRNA drugs.
- Jaden Weiss (4th year undergraduate - UoL) (currently: BSc student at U. of Lethbridge) May 2025 – Aug 2025
DFT mechanism of β -/ δ -hydrolysis nucleoside-protein adducts.
- Fatima Sadiq (2nd year undergraduate - UoL) (currently: BSc student at U. of Lethbridge) Jan 2025 – Aug 2025
DFT study of tRNA stacking interactions.
- Carl Day (3rd year undergraduate - UoL) (currently: BSc student at U. of Lethbridge) Sep 2024 – Dec 2024
TD-DFT and MD study of fluorescent NA probes.
- Haven Camillo (4th year undergraduate - UoG) (currently: MSc student at U. of Guelph) Jan 2024 – May 2024
Thermochemistry and ring-strain analysis of spring-loaded molecules.
- Noah King (4th year undergraduate/MSc - UoG) (currently: MSc student at U. of Guelph) Jan 2024 – Aug 2025
DFT study of palladium-catalyzed arylation of unsymmetrical bicycles.

- Alexander Wilson (4th year undergraduate - UoG) (currently: MSc student at U. of Guelph) Jan 2024 – May 2024
DFT study of rhodium-catalyzed rearrangements.
- Sebastian Karpinski (3rd year undergraduate - UoG) (currently: Medical school at U. of Ottawa) Apr 2021 – Sep 2021
DFT mechanism of ruthenium-catalyzed [2+2] cycloadditions.
- Krish K. Valluru (4th year undergraduate - UoG) (currently: Chemist at Eli Lilly) Sep 2020 – Apr 2021
DFT study of iridium-catalyzed hydroacylation of unsymmetrical bicycles.
- Fiona Bishop (4th year undergraduate - UoG) (currently: Chemist at Almac) Sep 2020 – Apr 2021
DFT mechanism of rhodium-catalyzed boronic acid ring-openings.
- Katrina Ruzicka (4th year undergraduate - UoG) (currently: PhD student at U. of Toronto) Sep 2020 – Apr 2021
DFT study of ruthenium-catalyzed [2+2] cycloadditions.
- Lina Ghulam (MSc - UoG) (currently: Pharmacist) Sep 2020 – Dec 2020
DFT mechanism of intermolecular acid-catalyzed ring-openings.

Technical Skills

- **Organic Synthesis:** Skilled in organic methodology development with a strong background in inert atmosphere techniques and glovebox chemistry, including reactions under Ar and H₂ gas. Crystallization of organic compounds for XRD.
- **Analytical Techniques:** Proficient in collecting and analyzing NMR, IR, and MS data. Skilled in purification methods including TLC and flash column chromatography.
- **Photochemistry:** Proficient in spectrophotometry and fluorometry. Skilled in fluorophore engineering and development.
- **Computational Chemistry:** Skilled in DFT and TD-DFT calculations for small molecules using **Gaussian** and **ORCA**. Experienced with molecular dynamics simulations of biomolecules using the **AMBER** suite. Familiar with Unix terminal environments and Bash and Python scripting for high-performance computing (Digital Research Alliance of Canada).